



Applied Statistics

Project Report

Project Title: Diabetes Prediction

Group No.		
Members:		
ID	Name	Email ID
232561	Afshan Waseem	232561@students.au.edu.pk

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Project Report: Diabetes Prediction Using Statistical Analysis

1. Abstract:

This study uses a diabetes prediction dataset and a wide variety of parametric and non-parametric statistical approaches. Understanding how different lifestyle and medical factors affect diabetes status, identifying important predictors, and visualizing the correlations between characteristics are the goals. R Studio is used in the project for graphical interpretation, statistical testing, and data cleansing.

2. Summary:

Data Source: We collect our data from the diabetic center located in Rawalakot Azad Kashmir.

Sample Size: Almost 150 patients fill the questionnaire..

Target variable:How do health-related factors vary between diabetic and non-diabetic patients?

Techniques Used: Shapiro-Wilk, T-tests, ANOVA, Wilcoxon, Kruskal-Wallis, Pearson & Spearman correlation, chi-square, and visualizations were among the methods employed.

Software used:R Studio

Result: Health factors such as Glucose, BMI, Insulin, and Age differ significantly between diabetic and non-diabetic patients. These factors also show meaningful relationships with each other and may serve as indicators or predictors of diabetes.

3. Introduction:

Diabetes is a long-term condition marked by high blood glucose levels. Problems can be considerably decreased with early detection. Using actual data, this study attempts to statistically evaluate a number of medical indications in order to forecast diabetes. In order to identify patterns and possible risk factors, both descriptive and inferential statistical methods are used.

4. Methodology:

- **Data Collection:**We collect our data from the diabetic center located in Rawalakot Azad Kashmir.
- **Data Preprocessing:**
 - Missing values check
 - Data type conversions
 - Grouping for categorical analysis (e.g., age, pregnancy)
- **Statistical Tests:**
 - Descriptive:** Central tendency and dispersion
 - Normality:** Shapiro-Wilk, Kolmogorov–Smirnov
 - Parametric:** T-test, ANOVA, Pearson correlation
- **Non-Parametric:** Wilcoxon, Kruskal-Wallis, Spearman correlation, Chi-Square

- **Visualization:** Boxplots, histograms, density plots, violin plots, bar graphs, pair plots

5. Analysis:

- **Descriptive Statistics:** Diabetic individuals show higher mean glucose and BMI values.
- **T-Tests/Wilcoxon:** Statistically significant differences found in glucose and insulin levels across diabetic statuses.
- **ANOVA/Kruskal-Wallis:** Age group has a significant impact on glucose levels.
- **Correlations:** Strong positive correlation between glucose and BMI; negative correlation with insulin in some groups.
- **Chi-Square:** Pregnancy count is significantly associated with diabetes.
- **Test used and Test Result:**
- **Shapiro-Wilk Test:**

Purpose: To confirm normality

Null Hypothesis: variables are normally distributed

P-value: most of the variables have $p\text{-value} < 0.05$

conclusion: Most variables not normally distributed

- **Kolmogorov–Smirnov Test:**

Purpose: Test for normality.

Null Hypothesis: Did variable follows a normal distribution or not.

P-value: most of the variables have $p\text{-value} < 0.05$.

Conclusion: Most variables not normally distributed

- **Independent T-Test:**

Purpose: Compare means between two groups

Null Hypothesis: Mean is equal in both groups

P-value: most of the variables have $p\text{-value} < 0.05$.

Conclusion: Diabetic and non-diabetic groups differ significantly.

- **Wilcoxon Test:**

Purpose: Compare medians when not normal

Null Hypothesis: Distribution is equal in both groups

P-value: mostly $p\text{-value} < 0.05$.

Conclusion: Confirms t-test results diabetic and non-diabetic groups differ significantly.

- **One-Way ANOVA:**

Purpose: Compare means across 3+ groups

Null Hypothesis: Mean glucose is equal across all age groups

P-value: $p\text{-value} < 0.05$.

Conclusion: Glucose levels differ across age groups

- **Kruskal-Wallis Test:**

Purpose: Non-parametric version of ANOVA

Null Hypothesis: Distribution of glucose is equal across age groups

P-value: $p\text{-value} < 0.05$.

Conclusion: Glucose levels differ across age groups

- **Pearson Correlation:**

Purpose: Test linear relationship

Null Hypothesis: No linear correlation between numeric variables

P-value: some $p\text{-value} > 0.05$

Conclusion: Strong linear relationships found (e.g., Glucose & Insulin)

- **Spearman Correlation:**

Purpose: Test monotonic (non-linear) relationships

Null Hypothesis: No linear correlation between numeric variables

P-value: some $p\text{-value} > 0.05$

Conclusion: Confirms relationships with less strict assumptions

- **Chi-square test:**

Purpose: Association between two categorical variables

Null Hypothesis: Pregnancy group and diabetes status are independent

P-value: some p-value > 0.05

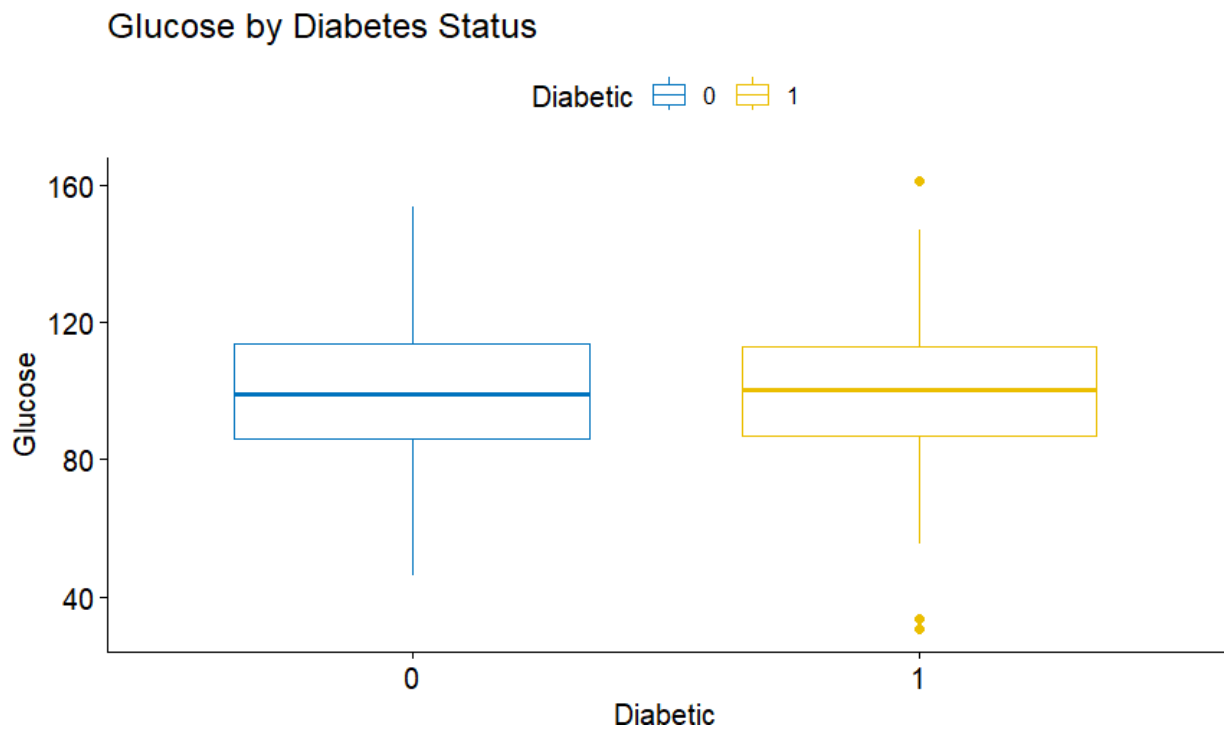
Conclusion: There is a statistically significant association between pregnancy group and diabetes status.

Visual Insights:

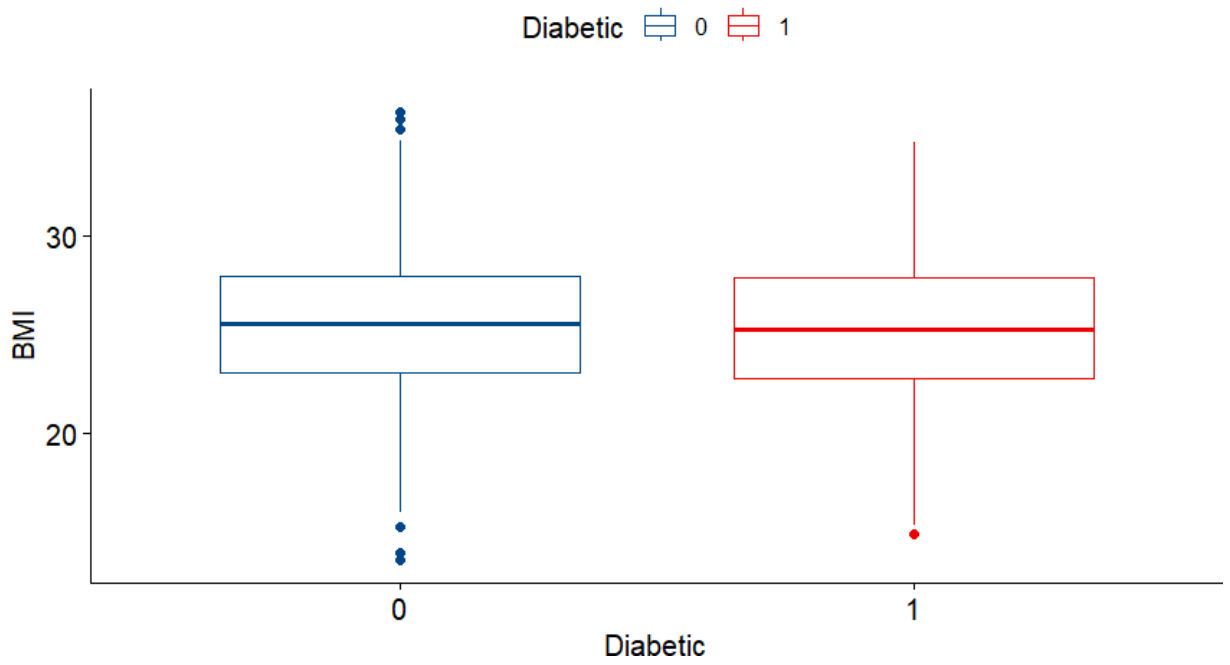
Violin plots show higher insulin variance among diabetics.

Bar plots indicate diabetes is more frequent in high pregnancy groups.

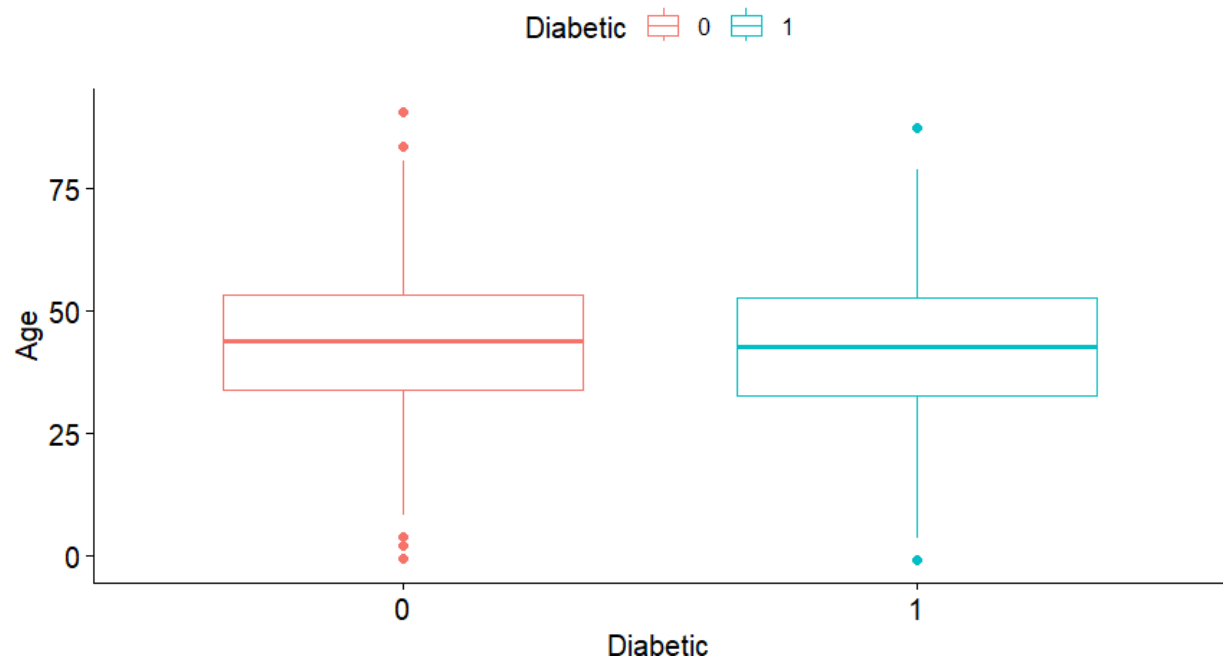
Density plots suggest higher age range skew in diabetic cases.

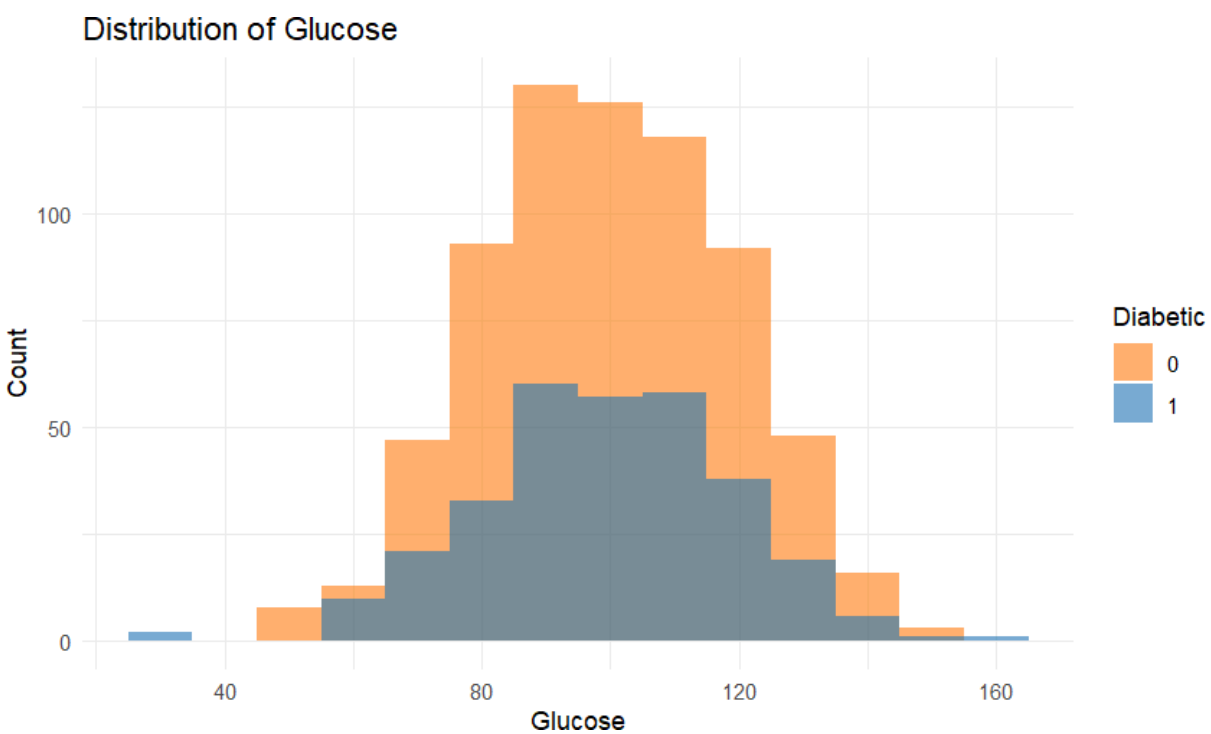
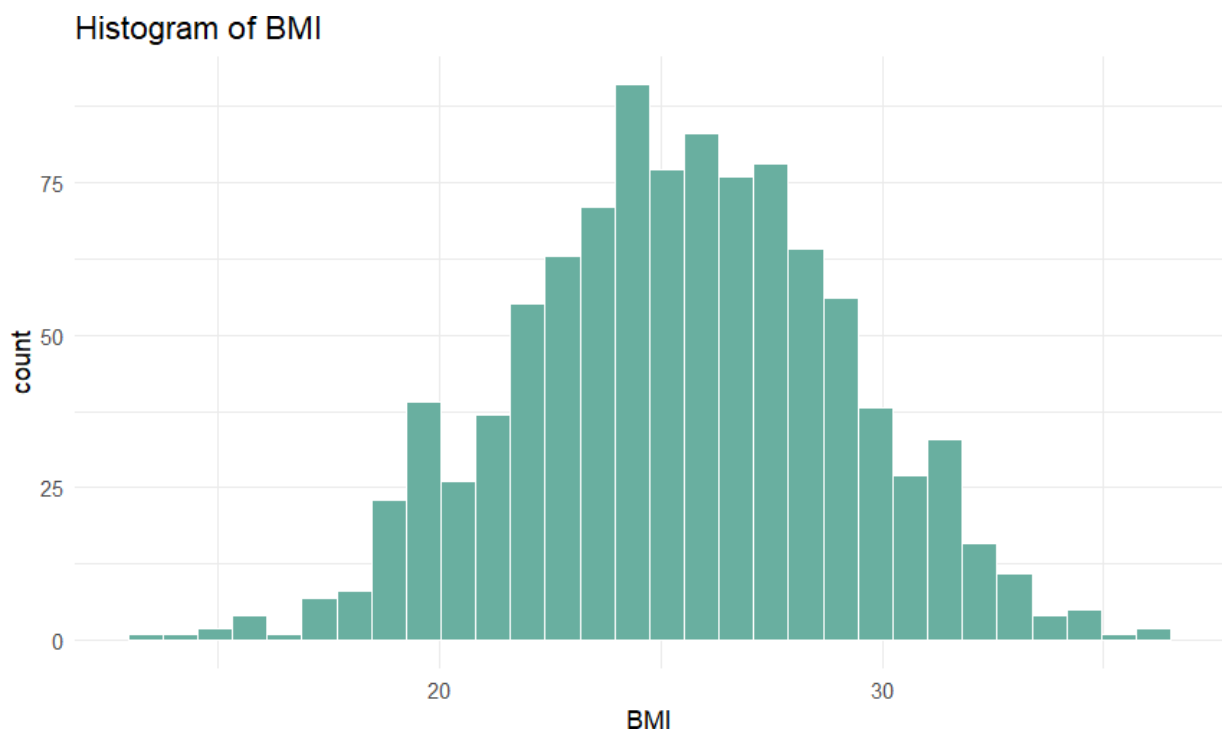


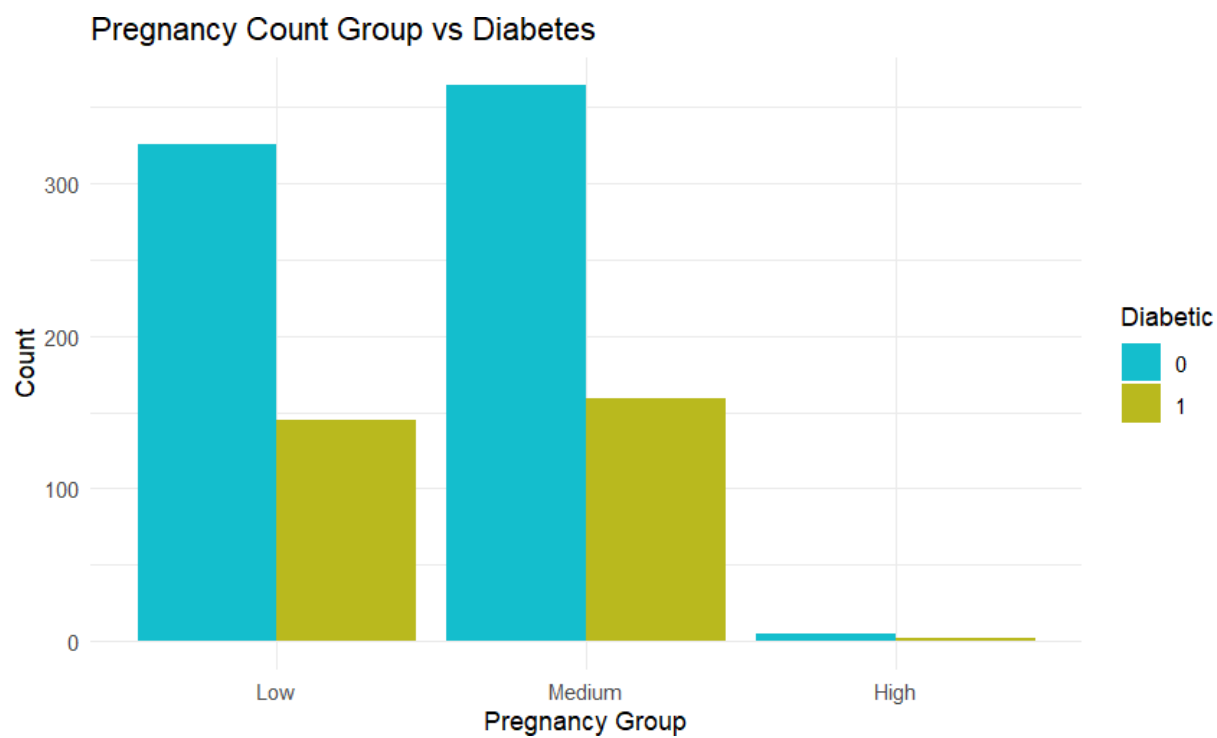
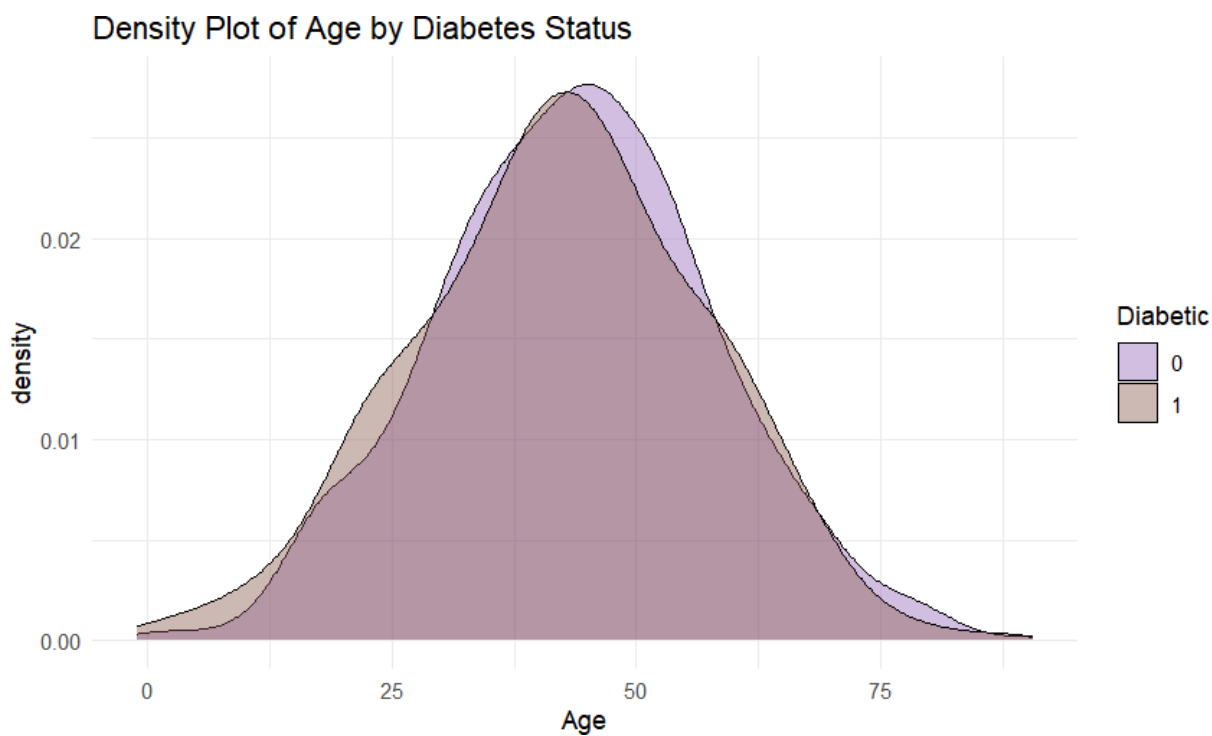
BMI by Diabetes Status



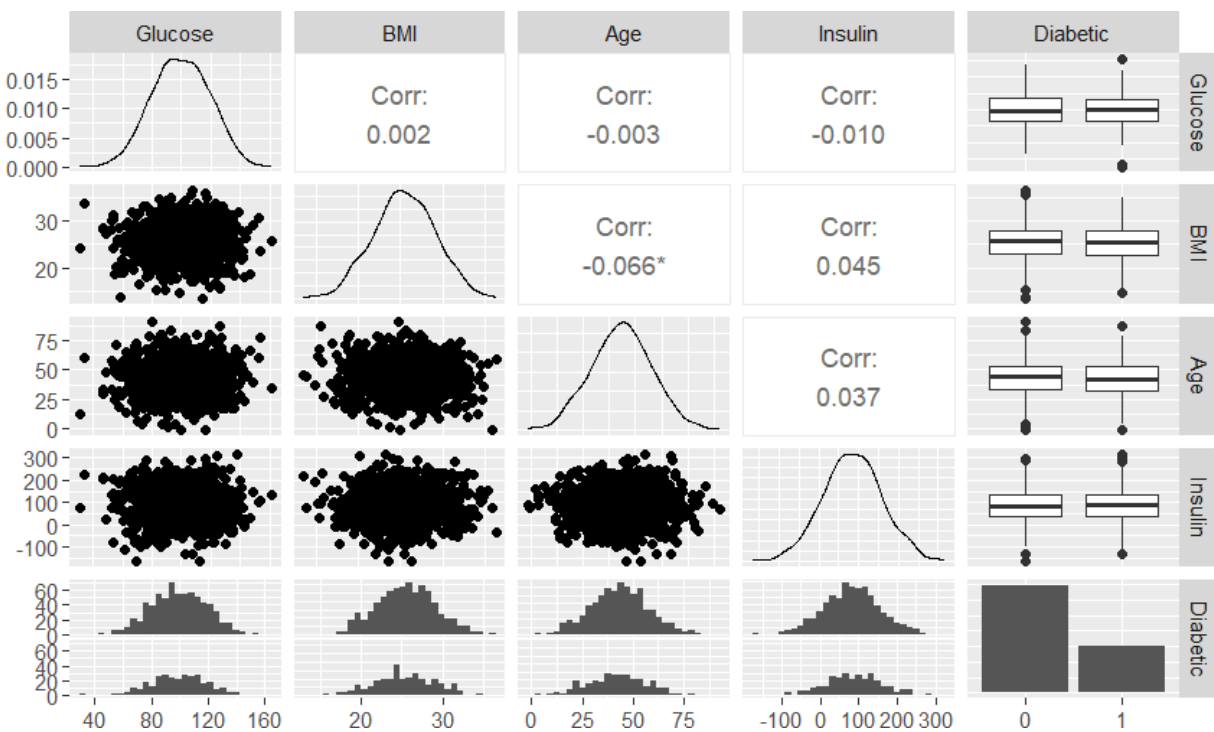
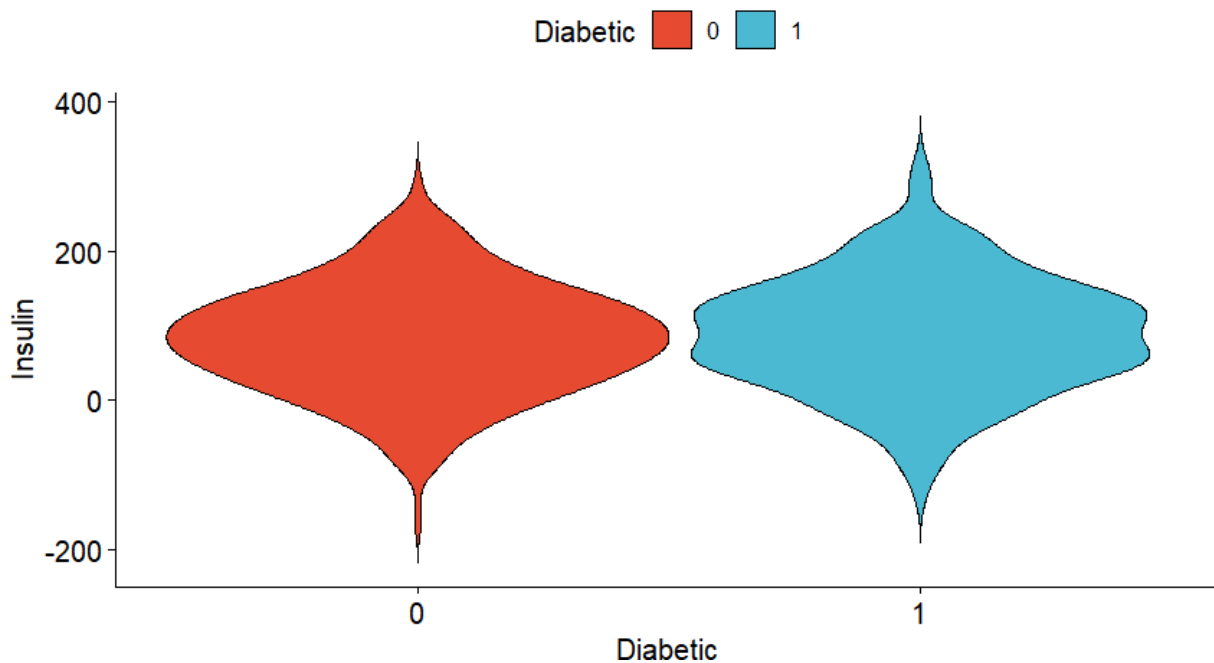
Age by Diabetes Status

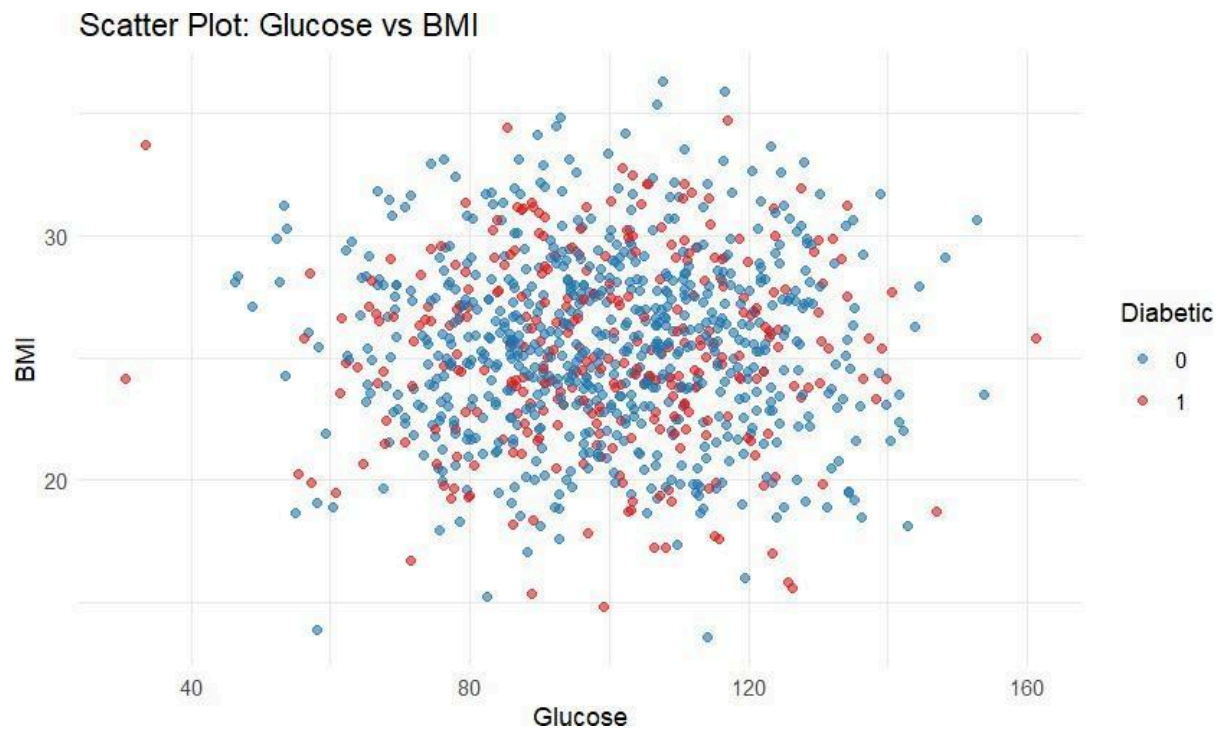
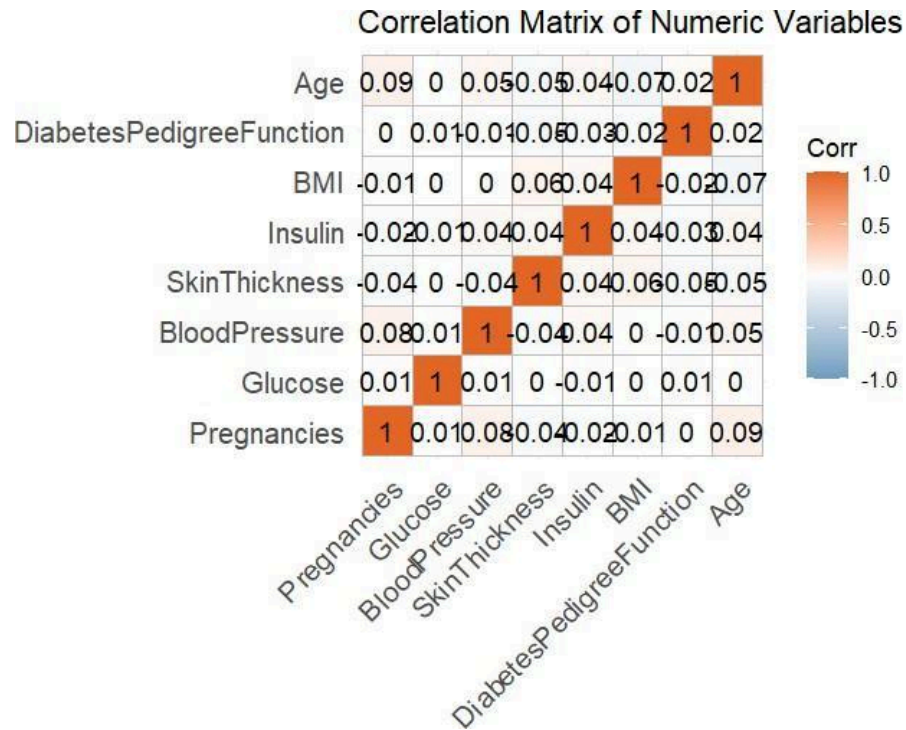


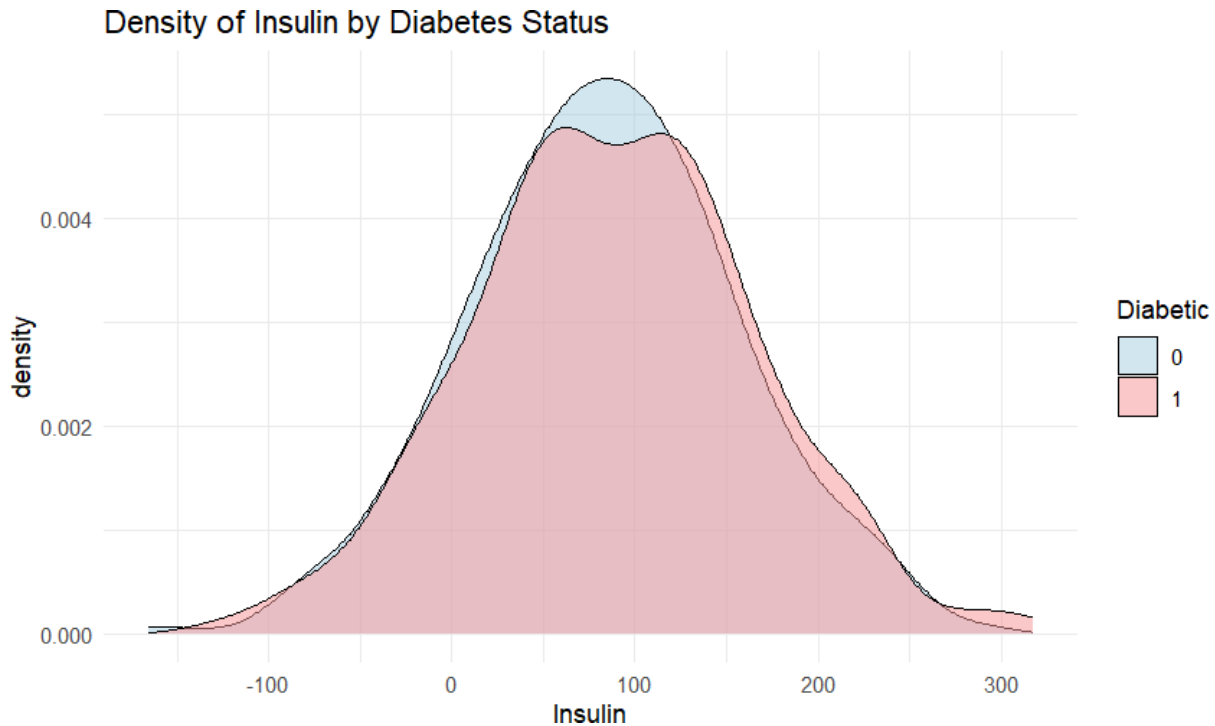




Insulin Levels by Diabetes Status







6. Conclusion:

The analysis of the dataset reveals significant differences between diabetic and non-diabetic individuals across several key health indicators. Diabetic individuals were found to have significantly higher levels of glucose, BMI, insulin, and age compared to non-diabetic individuals, as supported by both parametric (t-tests, ANOVA) and non-parametric (Wilcoxon, Kruskal-Wallis) tests. Tests for normality indicated that many variables were not normally distributed, justifying the use of non-parametric alternatives. Correlation analysis showed strong positive relationships between variables such as glucose and insulin, and BMI and insulin, suggesting that these factors often increase together. Additionally, a significant association was found between the number of pregnancies and diabetes status using the Chi-Square test, indicating that women with more pregnancies were more likely to be diabetic in this dataset. Overall, the results support the conclusion that health factors such as glucose levels, BMI, insulin, age, and pregnancy history play a meaningful role in distinguishing between diabetic and non-diabetic individuals.

7. Recommendations:

- To increase prediction accuracy, apply machine learning or logistic regression models.

- Encourage early screening for people with a high body mass index (BMI) or those who have had several pregnancies.
- Consider deploying an interactive questionnaire for real-time screening in community health programs.